

21. Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given that at least two are girls is :

माना प्रत्येक जन्म लेने वाले बच्चे का लड़का अथवा लड़की होना समसंभाव्य है। माना दो परिवारों में प्रत्येक में दो बच्चे हैं। यदि यह दिया गया है कि कम से कम दो बच्चे लड़कियाँ हैं, तो सभी बच्चों के लड़की होने की सप्रतिबंध प्रायिकता है :

- (A) $\frac{1}{11}$ (B) $\frac{1}{12}$ (C) $\frac{1}{10}$ (D) $\frac{1}{17}$

Ans. A

Sol. A = At least two girls

B = All girls

$$\begin{aligned} P\left(\frac{B}{A}\right) &= \frac{P(B \cap A)}{P(A)} \\ &= \frac{P(B)}{P(A)} = \frac{\left(\frac{1}{4}\right)^4}{1 - {}^4C_0\left(\frac{1}{2}\right)^4 - {}^4C_1\left(\frac{1}{2}\right)^4} \\ &= \frac{1}{16 - 1 - 4} = \frac{1}{11} \end{aligned}$$

22. ${}^{n-1}C_r = (k^2 - 8) {}^nC_{r+1}$ if and only if :

${}^{n-1}C_r = (k^2 - 8) {}^nC_{r+1}$ है यदि और केवल यदि :

- (A) $2\sqrt{2} < k \leq 3$ (B) $2\sqrt{3} < k < 3\sqrt{3}$
(C) $2\sqrt{3} < k \leq 3\sqrt{2}$ (D) $2\sqrt{2} < k < 2\sqrt{3}$

Ans. A

Sol. ${}^{n-1}C_r = (k^2 - 8) \frac{n}{r+1} {}^{n-1}C_r$

$$k^2 - 8 = \frac{r+1}{n}$$

$$0 < k^2 - 8 \leq 1$$

$$8 < k^2 \leq 9$$

$$2\sqrt{2} < k \leq 3 \text{ or } -3 \leq k < -2\sqrt{2}$$

23. If the equation $a|z|^2 + \overline{\alpha}z + \alpha\overline{z} + d = 0$ represents a circle where a, d are real constants, then which of the following condition is correct ?

(A) $\alpha = 0, a, d \in \mathbb{R}^+$

(B) $|\alpha|^2 - ad \neq 0$

(C) $|\alpha|^2 - ad > 0$ and $a \in \mathbb{R} - \{0\}$

(D) $|\alpha|^2 - ad \geq 0$ and $a \in \mathbb{R}$

यदि समीकरण $a|z|^2 + \overline{\alpha}z + \alpha\overline{z} + d = 0$ एक वृत्त को निरूपित करता है, जहाँ a, d वास्तविक अचर हैं, तो निम्न में से कौनसा सत्य है ?

(A) $\alpha = 0, a, d \in \mathbb{R}^+$

(B) $|\alpha|^2 - ad \neq 0$

(C) $|\alpha|^2 - ad > 0$ तथा $a \in \mathbb{R} - \{0\}$

(D) $|\alpha|^2 - ad \geq 0$ तथा $a \in \mathbb{R}$

Ans. C

Sol. $a|z|^2 + \overline{\alpha}z + \alpha\overline{z} + d = 0$

$\Rightarrow a|z|^2 + \alpha\overline{z} + \overline{\alpha}z + d = 0$

Let $z = x + iy; \alpha = p + iq$

$\Rightarrow a(x^2 + y^2) + (p + iq)(x - iy) + (p - iq)(x + iy) + d = 0$

$\Rightarrow a(x^2 + y^2) + 2px + 2qy + d = 0$

$\Rightarrow x^2 + y^2 + \frac{2p}{a}x + \frac{2q}{a}y + \frac{d}{a} = 0$

Represents a real circle if

$g^2 + f^2 - c > 0 \Rightarrow \frac{p^2}{a^2} + \frac{q^2}{a^2} - \frac{d}{a} > 0$

$\Rightarrow p^2 + q^2 > ad$

$\Rightarrow |\alpha|^2 - ad > 0 \text{ \& } a \neq 0.$

24. For all $z \in \mathbb{C}$ on the curve $C_1 : |z| = 4$, let the locus of the point $z + \frac{1}{z}$ be the curve C_2 . Then :

(A) the curves C_1 and C_2 intersect at 4 points

(B) the curve C_2 lies inside C_1

(C) the curve C_1 lies inside C_2

(D) the curves C_1 and C_2 intersect at 2 points

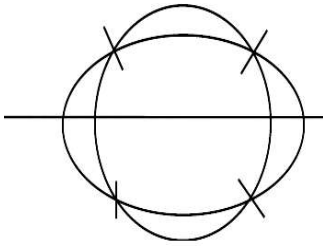
माना वक्र $C_1 : |z| = 4$ पर सभी $z \in \mathbb{C}$ के लिए बिंदु $z + \frac{1}{z}$ का बिंदुपथ वक्र C_2 है, तो :

- (A) वक्र C_1 तथा C_2 चार बिंदुओं पर मिलते हैं
 (B) वक्र C_2 , वक्र C_1 के अन्दर है
 (C) वक्र C_1 , वक्र C_2 के अन्दर है
 (D) वक्र C_1 तथा C_2 दो बिंदुओं पर मिलते हैं

Ans. A

Sol. $|z| = 4$ is a circle $x^2 + y^2 = 16$ and $z + \frac{1}{z}$ is an ellipse $\frac{x^2}{\left(\frac{17}{4}\right)^2} + \frac{y^2}{\left(\frac{15}{4}\right)^2} = 1$

$\therefore$ the curves C_1 and C_2 intersect at 4 points



25. Let α and β be the roots of the equation $x^2 + x + 1 = 0$. Then for $y \neq 0$ in $\mathbb{R}$, $\begin{vmatrix} y+1 & \alpha & \beta \\ \alpha & y+\beta & 1 \\ \beta & 1 & y+\alpha \end{vmatrix}$ is equal to:

माना α तथा β , समीकरण $x^2 + x + 1 = 0$ के मूल हैं, तो $\mathbb{R}$ में $y \neq 0$ के लिए $\begin{vmatrix} y+1 & \alpha & \beta \\ \alpha & y+\beta & 1 \\ \beta & 1 & y+\alpha \end{vmatrix}$ बराबर है :

- (A) $y(y^2 - 1)$ (B) $y^3 - 1$ (C) $y(y^2 - 3)$ (D) y^3

Ans. D

Sol. Let $\alpha = \omega$ and $\beta = \omega^2$ are roots of $x^2 + x + 1 = 0$

$\therefore \begin{vmatrix} y+\omega & 1 & \omega^2 \\ 1 & y+\omega^2 & \omega \\ \omega^2 & \omega & 1+y \end{vmatrix}$ operate $c_1 \rightarrow c_1 + c_2 + c_3$

$$y = \begin{vmatrix} 1 & 1 & \omega^2 \\ 1 & y + \omega^2 & \omega \\ 1 & \omega & 1 + y \end{vmatrix} \begin{pmatrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{pmatrix}$$

$$= y \begin{vmatrix} 1 & 1 & \omega^2 \\ 0 & y + \omega^2 - 1 & \omega - \omega^2 \\ 0 & \omega - 1 & 1 + y - \omega^2 \end{vmatrix}$$

$$= y \{ (y + \omega^2 - 1) (1 + y - \omega^2) - \omega(\omega - 1)(1 - \omega) \}$$

$$= y(y^2 - (\omega^2 - 1)^2) + y\omega(\omega - 1)^2$$

$$= y^3 + y(\omega - 1)^2 (\omega - (\omega + 1)^2) = y^3$$

26. If both the mean and the standard deviation of 50 observations $x_1, x_2, \dots, x_{50}$ are equal to 16, then the mean of $(x_1 - 4)^2, (x_2 - 4)^2, \dots, (x_{50} - 4)^2$ is :

यदि 50 प्रेक्षणों $x_1, x_2, \dots, x_{50}$ का माध्य तथा मानक विचलन दोनों 16 हैं, तो $(x_1 - 4)^2, (x_2 - 4)^2, \dots, (x_{50} - 4)^2$ का माध्य है:

- (A) 380 (B) 480 (C) 400 (D) 525

Ans. C

Sol. $\frac{x_1 + x_2 + \dots + x_{50}}{50} = 16$

$$16^2 = \frac{x_1^2 + x_2^2 + \dots + x_{50}^2}{50} - 16^2$$

$$2(16)^2 50 = x_1^2 + x_2^2 + \dots + x_{50}^2$$

$$\text{Required mean} = \frac{(x_1 - 4)^2 + (x_2 - 4)^2 + \dots + (x_{50} - 4)^2}{50}$$

$$\frac{16^2 (100) + 4^2 (50) - 8(16 \times 50)}{50}$$

$$= 16^2(2) + 16 - 8(16) = 400$$

27. The number of integral terms in the expansion of $\left(5^{\frac{1}{2}} + 7^{\frac{1}{8}}\right)^{1016}$ is :

$$\left(5^{\frac{1}{2}} + 7^{\frac{1}{8}}\right)^{1016} \text{ के प्रसार में पूर्णांकीय पदों की संख्या है।}$$

(A) 128

(B) 129

(C) 127

(D) 130

Ans. A

Sol. $T_{r+1} = 1016Cr.5^{\frac{1016-r}{2}} \cdot \frac{r}{7^8}$

for rational terms

$$\frac{1016-r}{2} \Rightarrow r = 0, \dots\dots\dots 1016$$

$$\frac{r}{8} = \text{integer} \Rightarrow r = 0, 8, 16, \dots\dots\dots 1016$$

Common $r = 0, 8, 16, \dots\dots\dots 1016$

$$1016 = 0 + (n-1) \times 8$$

$$10 = 8n$$

$$n = 128$$

28. If z be a complex number satisfying $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$, then $|z|$ cannot be :

यदि z एक ऐसी सम्मिश्र संख्या है जो $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$ को सन्तुष्ट करती है, तो $|z|$ नहीं हो सकता :

(A) $\sqrt{10}$

(B) $\sqrt{8}$

(C) $\sqrt{\frac{17}{2}}$

(D) $\sqrt{7}$

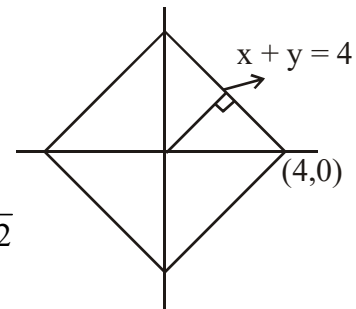
Ans. D

Sol. $z = x + iy$ $|z| = \sqrt{x^2 + y^2}$

$$\operatorname{Re}(z) = x \quad \operatorname{Im}(z) = y$$

$$|x| + |y| = 4$$

Minimum value of $|z|$ = distance of line $x + y = 4$ from origin = $2\sqrt{2}$



29. The minimum number of times one has to toss a fair coin so that the probability of observing at least one head is at least 90% is :

एक अनभिन्न सिक्के को कम से कम कितनी बार उछाला जाए ताकि कम से कम एक चित्त आने की प्रायिकता, कम से कम 90% हो?

(A) 5

(B) 2

(C) 4

(D) 3

Ans. C

Sol. $p = P(H) = \frac{1}{2}, q = 1 - p = \frac{1}{2}$

$$P(x \geq 1) \geq \frac{9}{19}$$

$$1 - P(x = 1) \geq \frac{9}{10}$$

$$1 - {}^nC_0 \left(\frac{1}{2}\right)^n \geq \frac{9}{10}$$

$$\frac{1}{2^n} \leq 1 - \frac{9}{10} \Rightarrow \frac{1}{2^n} \leq \frac{1}{10}$$

$$2^n \geq 10 \Rightarrow n \geq 4$$

$$\Rightarrow n_{\min} = 4$$

30. The probabilities of three events A, B and C are given by $P(A) = 0.6$, $P(B) = 0.4$ and $P(C) = 0.5$. If $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$, $P(A \cap B \cap C) = 0.2$, $P(B \cap C) = \beta$ and $P(A \cup B \cup C) = \alpha$, where $0.85 \leq \alpha \leq 0.95$, then β lies in the interval :

तीन घटनाओं A, B तथा C की प्रायिकताएँ $P(A) = 0.6$, $P(B) = 0.4$ तथा $P(C) = 0.5$ द्वारा दी गई हैं। यदि $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$, $P(A \cap B \cap C) = 0.2$, $P(B \cap C) = \beta$ तथा $P(A \cup B \cup C) = \alpha$, जहाँ $0.85 \leq \alpha \leq 0.95$, तो β निम्न में से किस अंतराल में है?

- (A) [0.25, 0.35] (B) [0.36, 0.40] (C) [0.20, 0.25] (D) [0.35, 0.36]

Ans. A

Sol. $P(A) = 0.6$ $P(B) = 0.4$

$$P(C) = 0.5 \quad P(A \cup B) = 0.8$$

$$P(A \cap C) = 0.3 \quad P(A \cap B \cap C) = 0.2$$

$$P(B \cap C) = \beta \quad P(A \cup B \cup C) = \alpha$$

$$0.85 \leq \alpha \leq 0.95$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.8 = 0.6 + 0.4 - P(A \cap B) \Rightarrow P(A \cap B) = 0.2$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$\alpha = 0.6 + 0.4 + 0.5 - 0.2 - \beta - 0.3 + 0.2$$

$$\alpha = 1.2 - \beta$$

$$\beta = 1.2 - \alpha$$

$$0.25 \leq \beta \leq 0.35$$

31. Let the mean and standard deviation of marks of class A of 100 students be respectively 40 and $\alpha (> 0)$, and the mean and standard deviation of marks of class B of n students be respectively 55 and $30 - \alpha$. If the mean and variance of the marks of the combined class of $100 + n$ students are respectively 50 and 350, then the sum of variance of classes A and B is :

माना 100 छात्रों की कक्षा A के छात्रों के अंकों के माध्य तथा मानक विचलन क्रमशः 40 तथा $\alpha (> 0)$ हैं तथा n छात्रों की कक्षा B के छात्रों के अंकों के माध्य तथा मानक विचलन क्रमशः 55 तथा $30 - \alpha$ है। यदि संयुक्त कक्षा के $100 + n$ छात्रों के अंकों के माध्य तथा प्रसरण क्रमशः 50 तथा 350 है, तो कक्षाओं A तथा B के प्रसरणों का योग है :

- (A) 650 (B) 500 (C) 450 (D) 900

Ans. B,D

Sol.	A	B	A+B
	$\bar{x}_1 = 40$	$\bar{x}_2 = 55$	$\bar{x} = 50$
	$\sigma_1 = \alpha$	$\sigma_2 = 30 - \alpha$	$\sigma_2 = 350$
	$n_1 = 100$	$n_2 = n$	$100 + n$

$$\bar{x} = \frac{100 \times 40 + 55n}{100 + n} = 50$$

$$\Rightarrow n = 200$$

$$\sigma_1^2 = \frac{\sum x_i^2}{100} - 40^2 = \alpha^2$$

$$\sigma_2^2 = \frac{\sum y_i^2}{200} - 55^2 = (30 - \alpha)^2$$

$$\sigma^2 = \frac{\sum x_i^2 + \sum y_i^2}{300} - 50^2 = 350^2$$

$$350^2 = \frac{(1600 + \alpha^2) \times 100 + [3025 + (30 - \alpha)^2] \times 200}{300} - 50^2$$

$$\text{on solving } \alpha^2 - 40\alpha + 300 = 0$$

$$\Rightarrow \alpha = 10 \text{ or } 30$$

$\alpha = 30$ is not possible

$$\sigma_1^2 + \sigma_2^2 = 10^2 + 20^2 = 500$$

32. Let the coefficients of three consecutive terms T_r , T_{r+1} and T_{r+2} in the binomial expansion of $(a + b)^{12}$ be in a

G.P. and let p be the number of all possible values of r . Let q be the sum of all rational terms in the binomial expansion of $(\sqrt[4]{3} + \sqrt[3]{4})^{12}$. Then $p + q$ is equal to :

माना कि $(a + b)^{12}$ के द्विपद प्रसार में तीन क्रमागत पदों T_r , T_{r+1} और T_{r+2} के गुणांक एक G.P. में हैं। माना कि r के सभी संभव मानों की संख्या p है। माना कि $(\sqrt[4]{3} + \sqrt[3]{4})^{12}$ के द्विपद प्रसार में सभी परिमेय पदों का योग q है। तब $p + q$ बराबर है :

- (A) 283 (B) 287 (C) 295 (D) 299

Ans. A

Sol. ${}^nC_{r-1}, {}^nC_r, {}^nC_{r+1} \Rightarrow \text{G.P.}$

$$n = 12$$

$$\Rightarrow {}^{12}C_{r-1}, {}^{12}C_r, {}^{12}C_{r+1} \Rightarrow \text{G.P.}$$

$$\Rightarrow ({}^{12}C_r)^2 = ({}^{12}C_{r-1}) ({}^{12}C_{r+1})$$

$$\Rightarrow \left(\frac{12!}{(12-r)!r!} \right) \left(\frac{12!}{(12-r)!r!} \right) = \left(\frac{12!}{(12-r+1)!(r-1)!} \right) \left(\frac{12!}{(12-r-1)!(r+1)!} \right)$$

$$\Rightarrow \frac{1}{r(12-r)} = \frac{1}{(12-r+1)(r+1)}$$

$$\Rightarrow (13-r)(r+1) = r(12-r)$$

$$\Rightarrow 13r + 13 - r^2 - r = 12r - r^2$$

$$13 = 0 \text{ Not possible}$$

0 values of r

$$\Rightarrow (\sqrt[4]{3} + \sqrt[3]{4})^{12} \Rightarrow \text{Rational terms}$$

$${}^{12}C_r (3)^{\left(\frac{12-r}{4}\right)} 4^{\left(\frac{r}{3}\right)}$$

$\Rightarrow r$ should be multiple of 4 & 3 means multiple of 12

$$r = 0 \text{ \& } 12$$

$$T_1 = ({}^{12}C_0) \left(3^{\left(\frac{12-0}{4}\right)} \right) (4^0) = 3^3 = 27$$

$$T_{13} = ({}^{12}C_{12}) (3^0) \left(4^{\frac{12}{3}} \right) = 4^4 = 256$$

$$27 + 256 = 283$$

33. Suppose $2 - p$, p , $2 - \alpha$, α are the coefficients of four consecutive terms in the expansion of $(1 + x)^n$. Then the value of $p^2 - \alpha^2 + 6\alpha + 2p$ equals :

माना $(1 + x)^n$ के प्रसार में चार क्रमागत पदों के गुणांक $2 - p$, p , $2 - \alpha$, α हैं। तो $p^2 - \alpha^2 + 6\alpha + 2p$ का मान बराबर है :

- (A) 8 (B) 10 (C) 6 (D) 4

Ans. Bonus

Sol. $2 - p$, p , $2 - \alpha$, α

Binomial coefficients are

nC_r , ${}^nC_{r+1}$, ${}^nC_{r+2}$, ${}^nC_{r+3}$ respectively

$$\Rightarrow {}^nC_r + {}^nC_{r+1} = 2$$

$$\Rightarrow {}^{n+1}C_{r+1} = 2 \quad \text{_____ (1)}$$

$$\text{Also, } {}^nC_{r+2} + {}^nC_{r+3} = 2$$

$$\Rightarrow {}^{n+1}C_{r+3} = 2 \quad \text{_____ (2)}$$

From (1) and (2)

$${}^{n+1}C_{r+1} = {}^{n+1}C_{r+3}$$

$$\Rightarrow 2r + 4 = n + 1$$

$$n = 2r + 3$$

$${}^{2r+4}C_{r+1} = 2$$

Data Inconsistent

34. Let α , β , γ and δ be the coefficients of x^7 , x^5 , x^3 and x respectively in the expansion of

$$\left(x + \sqrt{x^3 - 1}\right)^5 + \left(x - \sqrt{x^3 - 1}\right)^5, x > 1. \text{ If } u \text{ and } v \text{ satisfy the equations}$$

$$\alpha u + \beta v = 18$$

$$\gamma u + \delta v = 20$$

then $u + v$ equals :

माना α , β , γ तथा δ , $\left(x + \sqrt{x^3 - 1}\right)^5 + \left(x - \sqrt{x^3 - 1}\right)^5$, $x > 1$ के प्रसार में क्रमशः x^7 , x^5 , x^3 और x के गुणांक हैं। यदि u और

v , समीकरणों

$$\alpha u + \beta v = 18$$

$$\gamma u + \delta v = 20$$

को संतुष्ट करते हैं, तो $u + v$ बराबर है :

- (A) 8 (B) 4 (C) 5 (D) 3

Ans. C

Sol. $(x + \sqrt{x^3 - 1})^5 = {}^5C_0 x^5 + {}^5C_1 x^4 \sqrt{x^3 - 1} + {}^5C_2 x^3 (x^3 - 1) \dots\dots\dots$

$$(x + \sqrt{x^3 - 1})^5 = {}^5C_0 x^5 - {}^5C_1 x^4 \sqrt{x^3 - 1} + {}^5C_2 x^3 (x^3 - 1) \dots\dots\dots$$

$$= 2 \left({}^5C_0 x^5 + {}^5C_2 x^3 (x^3 - 1) + {}^5C_4 x^1 (x^3 - 1)^2 \right)$$

$$= 2 \left(x^5 + 10(x^6 - x^3) + 5x(x^6 + 1 - 2x^3) \right)$$

Now $\alpha = \text{coefficient of } x^7 = 10$

$\beta = \text{coefficient of } x^5 = 2$

$\gamma = \text{coefficient of } x^3 = -20$

$\delta = \text{coefficient of } x = 10$

Now $\alpha u + \beta v = 18$	$\gamma u + \delta v = 20$
$10u + 2v = 18$	$-20u + 10v = 20$
$5u + v = 9$	$-2u + v = 2$
$u = 1,$	$v = 4$

So $u + v = 5$

35. Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Let X denote the random variable of number of aces obtained in the two drawn cards. Then $P(X = 1) + P(X = 2)$ equals:

52 पत्तों की एक अच्छी प्रकार से फेंटी गई ताश की गड्डी में से, एक के बाद एक, दो पत्ते प्रतिस्थापना सहित निकाले गए। मान X, दोनों बार में प्राप्त इक्कों की संख्या को दर्शाने वाला यादृच्छिक चर है, तो $P(X = 1) + P(X = 2)$ बराबर है :

- (A) 24/169 (B) 25/169 (C) 52/169 (D) 49/169

Ans. B

Sol. $(\text{Ace})(\bar{\text{Ace}}) + (\bar{\text{Ace}})(\text{Ace}) + (\text{Ace})(\text{Ace})$

$$= \left(\frac{4}{52} \times \frac{48}{52} \right) \times 2 + \frac{4}{52} \times \frac{4}{52} = \frac{25}{169}$$

Method II

Binomial Probability Distribution $n = 2, p = 4/52$

Required Probability = $P(X = 1) + P(X = 2)$ where $P(X = r) = {}^nC_r p^r (1-p)^{n-r}$

36. For two non-zero complex numbers z_1 and z_2 , if $\operatorname{Re}(z_1 z_2) = 0$ and $\operatorname{Re}(z_1 + z_2) = 0$, then which of the following are possible?

- A. $\operatorname{Im}(z_1) > 0$ and $\operatorname{Im}(z_2) > 0$
- B. $\operatorname{Im}(z_1) < 0$ and $\operatorname{Im}(z_2) > 0$
- C. $\operatorname{Im}(z_1) > 0$ and $\operatorname{Im}(z_2) < 0$
- D. $\operatorname{Im}(z_1) < 0$ and $\operatorname{Im}(z_2) < 0$

Choose the correct answer from the options given below :

- (A) B and C (B) B and D (C) A and B (D) A and C

z_1 व z_2 दो शून्येतर सम्मिश्र संख्याएँ हैं, यदि $\operatorname{Re}(z_1 z_2) = 0$ और $\operatorname{Re}(z_1 + z_2) = 0$ है, तब

- A. $\operatorname{Im}(z_1) > 0$ एवं $\operatorname{Im}(z_2) > 0$
- B. $\operatorname{Im}(z_1) < 0$ एवं $\operatorname{Im}(z_2) > 0$
- C. $\operatorname{Im}(z_1) > 0$ एवं $\operatorname{Im}(z_2) < 0$
- D. $\operatorname{Im}(z_1) < 0$ एवं $\operatorname{Im}(z_2) < 0$

नीचे दिये गये विकल्पों से सही उत्तर चुनिए :

- (A) B तथा C (B) B तथा D (C) A तथा B (D) A तथा C

Ans. A

Sol. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$
 $\therefore \operatorname{Re}(z_1 z_2) = 0$ and $\operatorname{Re}(z_1 + z_2) = 0$,

$$x_1 x_2 - y_1 y_2 = 0 \quad \dots(i)$$

$$x_1 + x_2 = 0 \quad \dots(ii)$$

$$x_1^2 + y_1 y_2 = 0$$

$$y_1 y_2 = -x_1^2$$

$\Rightarrow \operatorname{Im}(z_1)$ and $\operatorname{Im}(z_2)$ are of opposite sign.

37. If the equation, $x^2 + bx + 45 = 0$ ($b \in \mathbb{R}$) has conjugate complex roots and they satisfy $|z + 1| = 2\sqrt{10}$, then:

यदि समीकरण $x^2 + bx + 45 = 0$ ($b \in \mathbb{R}$) के संयुग्मी सम्मिश्र मूल हैं, जो $|z + 1| = 2\sqrt{10}$ को संतुष्ट करते हैं, तो :

- (A) $b^2 + b = 72$ (B) $b^2 - b = 30$ (C) $b^2 + b = 12$ (D) $b^2 - b = 42$

Ans. B

Sol. Let $z = \alpha \pm i\beta$ be roots of the

$$\text{equation } x^2 + bx + 45 = 0$$

$$\text{sum of roots} = 2d = -b$$

$$\text{Product of roots} = \alpha^2 + \beta^2 = 45 \quad \dots\dots\dots(1)$$

$$|z + 1| = 2\sqrt{10}$$

$$|\alpha \pm i\beta + 1| = 2\sqrt{10}$$

$$(\alpha + 1)^2 + \beta^2 = 40 \quad \dots\dots\dots(2)$$

from (1) and (2)

$$2\alpha + 1 = 5$$

$$\alpha = -3$$

$$b = 6$$

38. A person throws two fair dice. He wins Rs. 15 for throwing a doublet (same numbers on the two dice), wins Rs. 12 when the throw results in the sum of 9, and loses Rs. 6 for any other outcome on the throw. Then the expected gain/ loss (in Rs.) of the person is

- (A) 1/2 loss (B) 2 gain (C) 1/2 gain (D) 1/4 loss

एक व्यक्ति दो न्याय्य (fair) पासे उछालता है। एक द्विक (दोनों पासों पर एक ही संख्या) आने पर वह रुपये 15 जीतता है, दोनों पासों पर आए अंकों का योग 9 होने पर रुपये 12 जीतता है तथा किसी अन्य परिणाम (outcome) पर रुपये 6 हारता है, तो उस व्यक्ति का प्रत्याशित (expected) लाभ/हानि (रुपयें में) है :

- (A) 1/2 हानि (B) 2 लाभ (C) 1/2 लाभ (D) 1/4 हानि

Ans. A

Sol. Let X be the random variable which denote the Rs gained by the person.

$$P(X = 15) = \frac{6}{36} = \frac{1}{6} \quad \left\{ \begin{array}{l} \text{Total cases} = 36 \\ \text{favourable cases are} \\ (1,1), (2,2), (3,3), (4,4), \\ (5,5), (6,6) \end{array} \right\}$$

$$P(X = 12) = \frac{4}{36} = \frac{1}{9} \quad \left\{ \begin{array}{l} \text{Favourable cases are} \\ (6,3), (5,4), (4,5), (3,6) \end{array} \right\}$$

$$P(X = -6) = \frac{26}{36} = \frac{13}{18}$$

X	15	12	-6
P(X)	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{13}{18}$
X · P(X)	$\frac{5}{2}$	$\frac{4}{3}$	$-\frac{13}{3}$

$$E(X) = \sum X \cdot P(X) = \frac{5}{2} + \frac{4}{3} - \frac{13}{3} = -\frac{1}{2}$$

39. If α and β be the roots of the equation $x^2 - 2x + 2 = 0$ then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$ is

यदि समीकरण $x^2 - 2x + 2 = 0$ के मूल α तथा β हैं, तो n का न्यूनतम मान, जिसके लिए $\left(\frac{\alpha}{\beta}\right)^n = 1$ है, है :

- (A) 4 (B) 5 (C) 3 (D) 2

Ans. A

Sol. $x^2 - 2x + 2 = 0$

roots of this equation are $\frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i$

$$\text{Then } \frac{\alpha}{\beta} = \frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = i$$

$$\text{or } \frac{\alpha}{\beta} = \frac{1-i}{1+i} = \frac{(1-i)^2}{1-i^2} = -i$$

$$\text{So, } \frac{\alpha}{\beta} = \pm i$$

$$\text{Now, } \left(\frac{\alpha}{\beta}\right)^n = 1 \quad \therefore i^n = 1$$

"n must be a multiple of 4 minimum value of $n = 4$

40. The number of 6 digit numbers that can be formed using the digits 0, 1, 2, 5, 7 and 9 which are divisible by 11 and no digit is repeated, is :

अंकों (digits) 0, 1, 2, 5, 7 तथा 9 के प्रयोग से छः अंकों वाली ऐसी संख्याओं, जो 11 से भाज्य हों तथा जिनमें कोई भी अंक दोबारा न आए, की संख्या है :

- (A) 72 (B) 48 (C) 60 (D) 36

Ans. C

Sol.

a_1	a_2	a_3	a_4	a_5	a_6
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 digit 0, 1, 2, 5, 7, 9

$$(a_1 + a_3 + a_5) - (a_2 + a_4 + a_6) = 11K$$

so (1, 2, 9) (0, 5, 7)

Now number of ways to arranging them

$$= 3! \times 3! + 3! \times 2 \times 2$$

$$= 6 \times 6 + 6 \times 4$$

$$= 6 \times 10$$

$$= 60$$